Tools are needed to reduce the time and cost to design superconductor-based circuits, potentially revolutionising the computer and electronics industry. Therefore, to advance the science of superconducting, researchers have begun to build electronic design automation and technology computer-aided design (TCAD) tools that will make it simpler to blueprint circuits based on superconducting materials.

The programme's short-term goal is to significantly reduce design time and increase the reliability of designs for complex circuits made of superconducting materials. The longer-term goal is to achieve a full, integrated design automation chain for digital-analogue hybrid circuits. By making it faster and easier to design such circuits, researchers intend to spur the widespread adoption of superconductor technologies.

Ultimately, the availability of design tools may foster very-large-scale integrated design of superconducting electronics, which can include millions or even hundreds of millions of components on a single chip.

## Building tools

Physicists found a way to control electrical transport through a superconducting material by building a device within it. Called Josephson junction, the device is analogous in function to the transistor in semiconductor electronics. It's composed of two superconducting electrodes separated by about one nanometer—a billionth of a meter.

Circuits built from Josephson junctions are called SQUIDS, and used for detectors of extremely small magnetic fields, that are more than ten billion times smaller than Earth's. However, these devices require low temperatures to operate, typically just 4 degrees above absolute zero. This requires intricate and costly cooling systems.

Nearly three decades have passed since the discovery of the first high-temperature superconductor, but progress in building electronic devices using these materials has been very slow, because process control at the sub-10-nanometre scale is required to make high-quality Josephson junctions out of these materials.

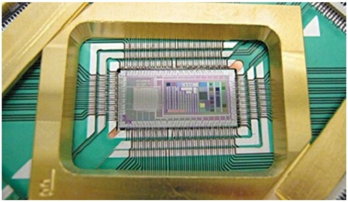
Further developments led to ceramic materials that become superconducting—that is, lose all resistance to electricity—at temperatures that can be easily achieved in the laboratory with liquid nitrogen. Discovery of high-temperature superconductivity has set off an intense effort to develop new kinds of electronics and other devices with these new materials.

In yet another development, physicists have found a new way to control the transport of electrical currents through high-temperature superconductors. Their achievement paves the way for the development of sophisticated electronic devices that are capable of allowing scientists or clinicians to non-invasively measure tiny magnetic fields in the heart or brain, and improve satellite communications.

This new approach will have a significant and far-reaching impact in medicine, physics, materials science and satellite communications. It will enable development of a new generation of superconducting electronics covering a wide spectrum, ranging from highly sensitive magnetometers for biomagnetic measurements of the human body to large-scale arrays for wideband satellite communications. In basic science, it could contribute to unravelling the mysteries of unconventional super-

### WINNERS OF THE NOBLE PRIZE IN PHYSICS FOR SUPERCONDUCTIVITY

| Recipients | Year of award | Discovery/research |
|---|---|---|
| Heike Kamerlingh Onnes | 1913 | For his investigations on the properties of matter at low temperatures, which led, inter alia, to the production of liquid helium |
| John Bardeen, Leon Neil Cooper and John Robert Schrieffer | 1972 | For their jointly developed theory of superconductivity, usually called the BCS-theory |
| Leo Esaki, Ivar Giaever and Brian D. Josephson | 1973 | For their experimental discoveries regarding tunneling phenomena in semiconductors and superconductors, respectively, and for theoretical predictions of the properties of a supercurrent through a tunnel barrier, in particular phenomena that are generally known as the Josephson effects |
| J. Georg Bednorz and K. Alexander Müller | 1987 | For their important breakthrough in the discovery of superconductivity in ceramic materials |
| Alexei A. Abrikosov, Vitaly L. Ginzburg and Anthony J. Leggett | 2003 | For pioneering contributions to the theory of superconductors and superfluids |



*Superconducting electronics*